

41680 F23 Introduction to advanced materials (exam)

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

- Der er altid netop ét svar som er mere rigtigt end de andre
- Studerende kan kun vælge ét svar per spørgsmål
- Hvert rigtigt svar giver 1 point
- Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"

- There is always only one correct answer – a response that is more correct than the rest
- Students are only able to select one answer per question
- Every correct answer corresponds to 1 point
- Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

Page 1

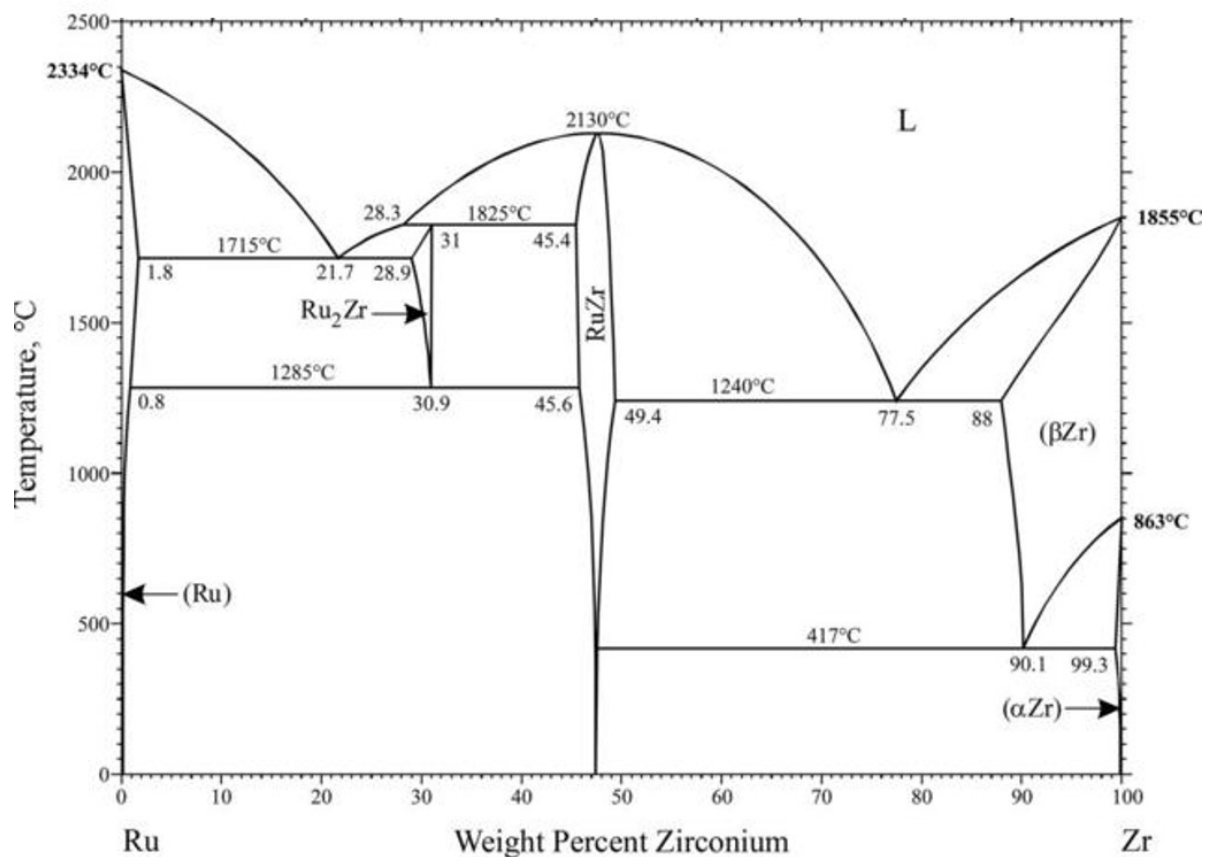
Consider the semiconductor Si doped with As. Which of the following statements about the carriers of electricity is correct?

- ☒ At low temperatures, the carrier densities are given by the doping concentration, at high temperatures, they become similar to that of undoped Si.
- ☐ The carrier densities are approximately constant at all temperatures.
- ☐ At low temperatures, there is an equal amount of electrons in the conduction band and holes in the valence band
- ☐ At low temperatures, there is an equal amount of electrons in the valence band and holes in the conduction band.
-

A copper smith hammers a piece of α -brass (a single phase Cu-Zn alloy with less than 38 wt.% Zn) into a new shape at room temperature. Why does the material become stronger?

- ☒ The grain size becomes reduced.
- ☐ More dislocations are formed.
- ☐ The atoms are squeezed together.
- ☐ Cu and Zn atoms are mixed in a better way.

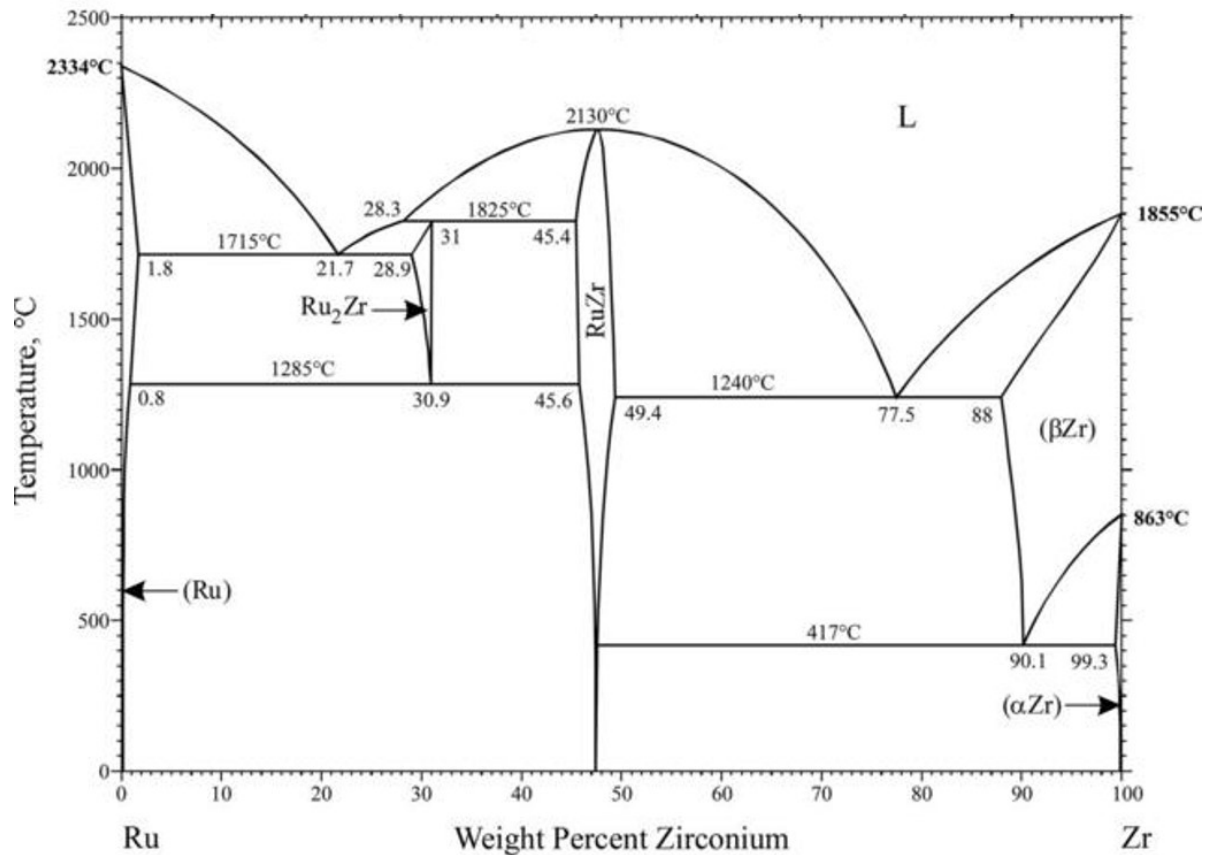
The figure shows the phase diagram of the binary alloy system Ru-Zr.



A melt of a ruthenium-zirconium alloy with **77.5 wt.% Zr** is cooled slowly from 2500 °C to room temperature. What is the mass of RuZr at room temperature for a melt of 6.1 kg?

- ☒ About 3.5 kg.
- ☐ About 2.6 kg.
- ☐ About 4.5 kg.
- ☐ About 1.6 kg.

The figure shows the phase diagram of the binary alloy system Ru-Zr.



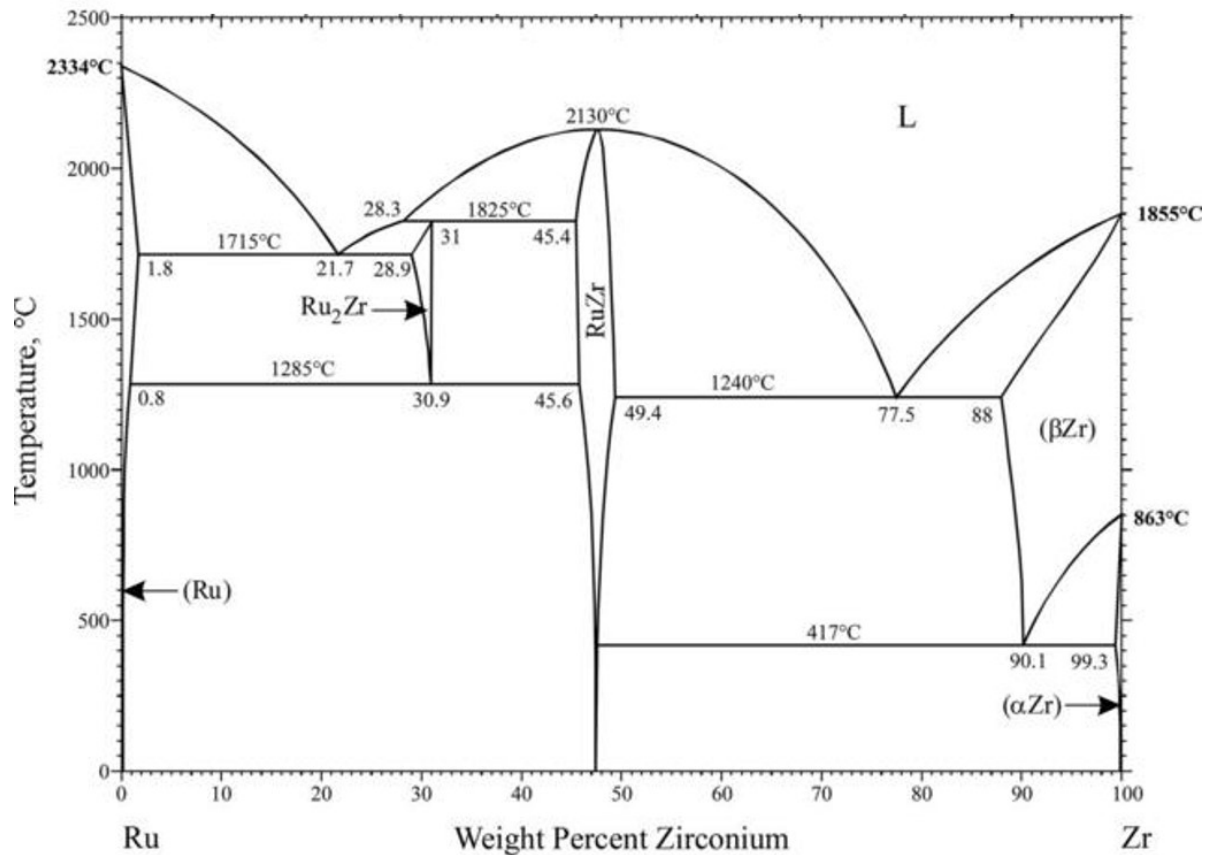
Where does a eutectoid transformation occur in the binary system ruthenium-zirconium?

- ☐ No eutectoid transformation occurs in the system.
- ☐ A eutectoid transformation occurs at about 1825 °C for 31 wt.% Zr.
- ☒ A eutectoid transformation occurs at about 417 °C for 90.1 wt.% Zr.
- ☐ A eutectoid transformation occurs at about 1715 °C for 21.7 wt.% Zr.

Where are foreign atoms not to be expected, when a pure metal is alloyed with a second element of slightly smaller atomic size?

- ☐ At vacancies.
- ☐ At dislocations.
- ☒ On regular lattice sites.
- ☐ On interstitial sites.
- ☐ At grain boundaries.

The figure shows the phase diagram of the binary alloy system Ru-Zr.



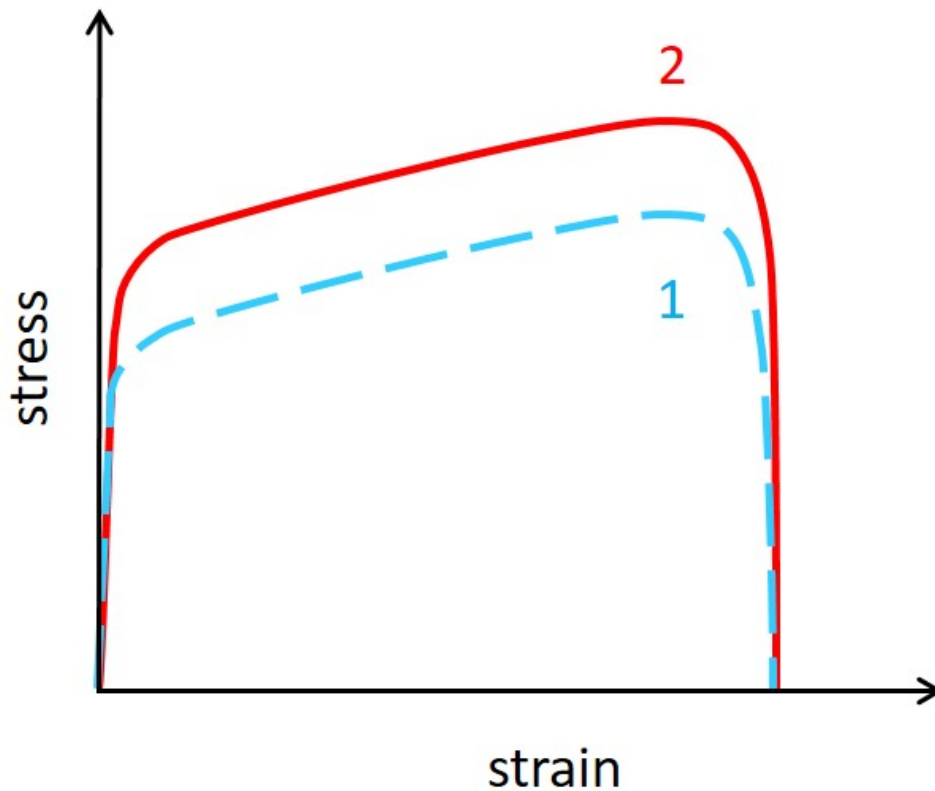
What is the maximum solubility of Ru in solid zirconium?

- ☐ 88 wt.% Ru
- ☐ 99.3 wt.% Ru
- ☒ 0.7 wt.% Ru
- ☐ 12 wt.% Ru

Which of the following statements about phonons is not correct?

- ☐ Phonons are related to lattice vibrations in crystalline materials.
- ☐ Phonons transport electricity well in crystalline materials.
- ☒ Phonons are a collective motion of atoms, giving rise to waves.
- ☐ Phonons transport heat well in crystalline materials.

The stress strain diagram shows two flow curves obtained for two different metallic specimens (1 and 2) at room temperature under the same conditions.



Which of the two metallic specimens should be selected for a light and stiff pillar of a given length?

- ☒ Both specimens are equally suited, if both specimen have the same mass density.
 - ☐ Specimen 1, if both specimens have the same mass density.
 - ☐ Specimen 2, if both specimens have the same mass density.
 - ☐ The specimen with lower mass density, independent of the flow curve.
-

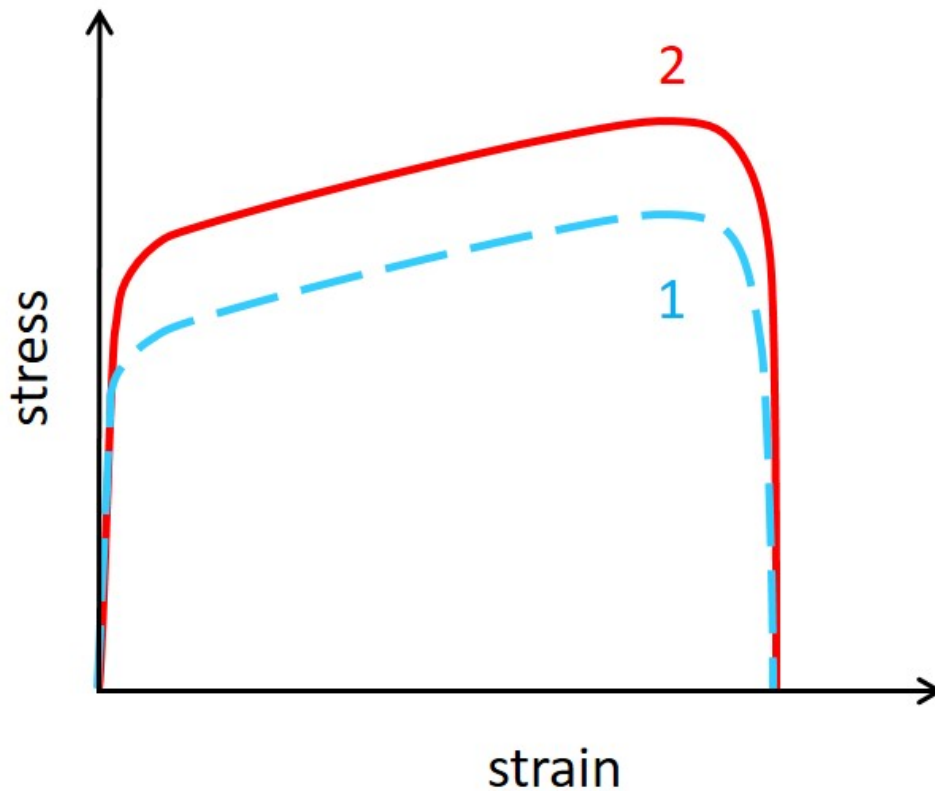
A piece of jewelry is manufactured as layered composite from several thin foils made of two different metals (copper and gold). For economic reasons, the volume fraction of gold is only 15 %. The table summarizes mechanical properties of both metals.

Metal	Young's modulus / GPa	Yield strength / MPa	Tensile strength / MPa	Ductility / %
Cu	130	130	283	17
Au	78.5	184	199	3.5

What is the Young's modulus of the layered composite under compression perpendicular to the foils?

- ☐ Less than 85 GPa.
 - ☐ At least 125 GPa.
 - ☒ At least 105 GPa, but less than 120 GPa.
 - ☐ At least 120 GPa, but less than 125 GPa.
 - ☐ At least 90 GPa, but less than 105 GPa.
 - ☐ At least 85 GPa, but less than 90 GPa.
-

The stress strain diagram shows two flow curves obtained for two different metallic specimens (1 and 2) at room temperature under the same conditions.



What could be the difference between the two metallic specimens?

- ☐ Both specimens may have the same composition, but they have been work-hardened to a different extend.
- ☐ Both specimens may have the same composition, but one must have larger grains and must also have been work-hardened to a larger extend than the one with smaller grain size.
- ☐ The two specimens must obviously have a different chemical composition.

☒ Both specimens may have the same composition, but a different grain size.

Stiffness and toughness of a metal depend on temperature. When a body-centered cubic metal is cooled to extremely low temperatures, ...

- ☒ ... both, its stiffness and its toughness, increase.
 - ☐ ... its toughness increases, but not its stiffness.
 - ☐ ... its stiffness increases, but not its toughness.
 - ☐ ... neither its stiffness, nor its toughness increases.
-

Consider steady state transport of heat from a sphere with a diameter of 2 m and a uniform wall thickness of 15 cm. The wall has a thermal conductivity of 0.2 W/(m·K). How much power is required to be generated inside the sphere, if a constant temperature difference has to be maintained such that the inside is 15 K warmer than the outside of the sphere?

- ☐ About 1.3 MJ/h
- ☐ About 130 MJ/h
- ☐ About 55 MJ/h

☒ About 900 kJ/h

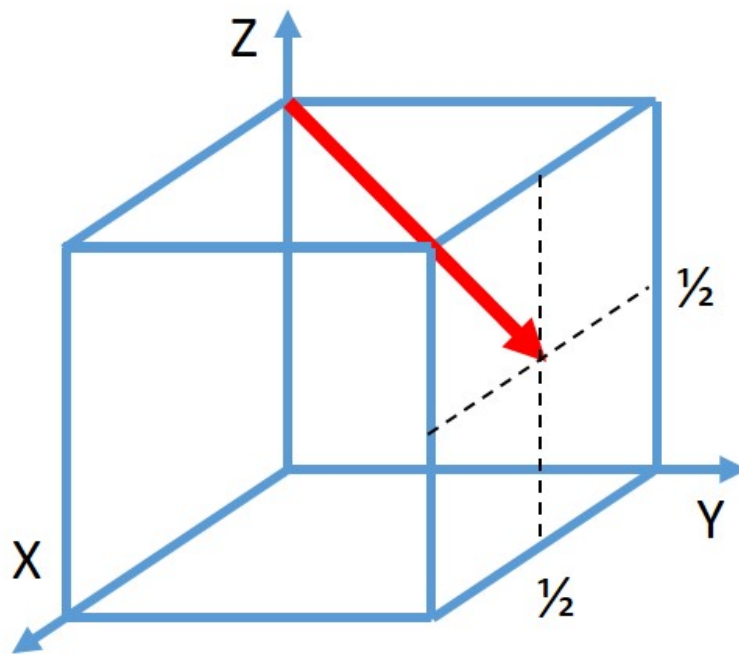
The picture shows plastic beverage cans at room temperature.



What can be concluded about the polymer the cans are made of?

- ☐ The polymer must be semi-crystalline and its glass transition temperature must be above room temperature.
 - ☐ The polymer must be semi-crystalline and its glass transition temperature must be below room temperature.
 - ☒ The polymer cannot be semi-crystalline and its glass transition temperature must be above room temperature.
 - ☐ The polymer cannot be semi-crystalline and its glass transition temperature must be below room temperature.
-

What is the proper notation of the particular crystallographic direction shown in the unit cell in the figure?



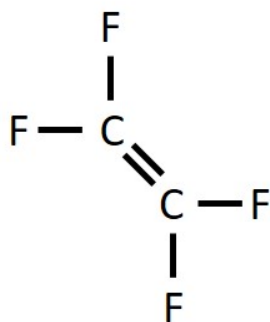
☐ $[\bar{1}\bar{2}1]$

☐ $[21\bar{2}]$

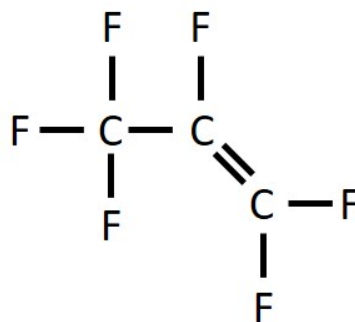
☐ $[\bar{2}\bar{1}2]$

☒ $[12\bar{1}]$

Fluorinated ethylene propylene (FEP) is a copolymer of tetrafluoroethylene and hexafluoropropylene, while polytetrafluoroethylene (PTFE) is solely build from tetrafluoroethylene. The figure shows the monomers.



Tetrafluoroethylene

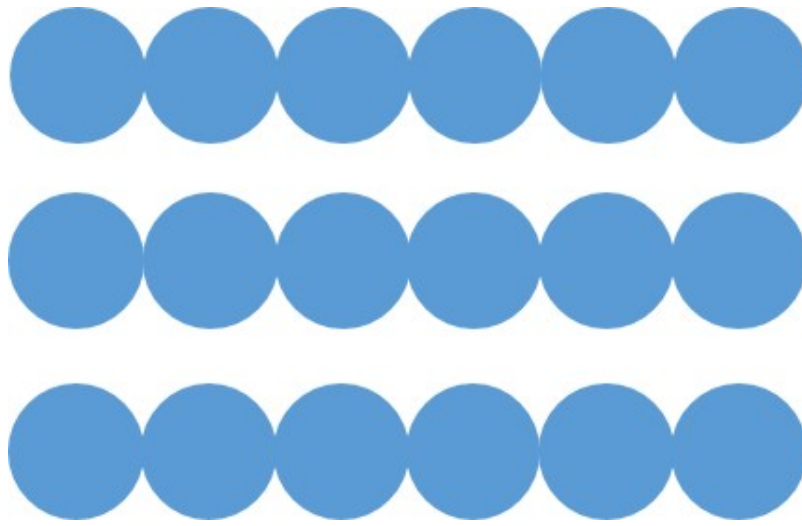


Hexafluoropropylene

Which statement is correct when comparing two batches of FEP and PTFE with the same degree of polymerization?

- ☐ FEP has not a longer straightened chain length, but a higher molar mass than PTFE.
 - ☐ FEP has neither a longer straightened chain length, nor a higher molar mass than PTFE.
 - ☒ FEP has a longer straightened chain length and a higher molar mass than PTFE.
 - ☐ FEP has a longer straightened chain length, but not a higher molar mass than PTFE.
-

Which crystallographic plane is shown in the figure?



☐ A {110} plane in a simple cubic lattice.

☒ A {110} plane in a face-centered cubic lattice.

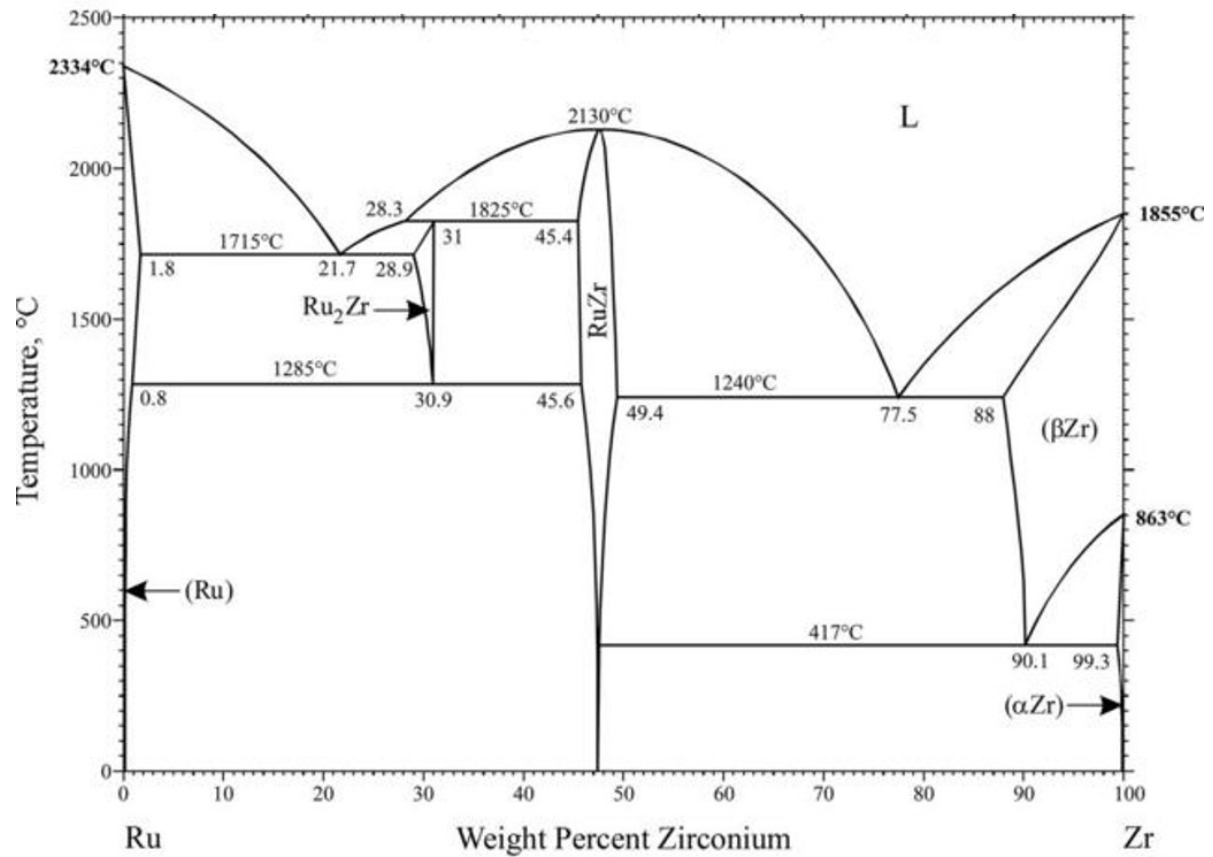
☐ A {100} plane in a simple cubic lattice.

☐ A {100} plane in a face-centered cubic lattice.

☐ A {110} plane in a body-centered cubic lattice.

☐ A {100} plane in a body-centered cubic lattice.

The figure shows the phase diagram of the binary alloy system Ru-Zr.



A melt of a ruthenium-zirconium alloy with **38 wt.% Zr** is cooled slowly from 2500 °C to room temperature. Which is the first solid phase formed?

- ☐ (Ru)
- ☐ (Ru) + L
- ☐ Ru₂Zr
- ☐ Ru₂Zr + L
- ☐ RuZr + L

☒ RuZr

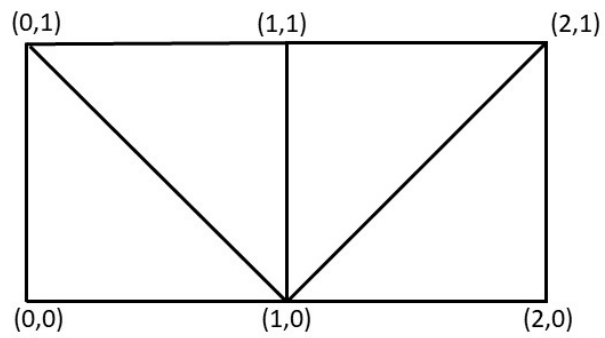
The fictive element Melinium (Ml) forms an oxide and a sulfide, both with cubic crystal structure. The table provides some properties of the elements.

Element	Valence	Elektronegativity (Pauling)	Atomic radius / pm	Ionic radius / nm
Ml	+2	1.7	149	95
O	-2	3.5	60	140
S	-2	2.5	106	184

Which statement concerning the compounds MlO and MlS are correct?

- ☐ Both compounds have different crystal structures and their fraction of ionic bonding is at least 50 %.
- ☐ Both compounds have the same crystal structure and their fraction of ionic bonding is at least 50 %.
- ☐ Both compounds have different crystal structures and at least one of the compounds has a fraction of ionic bonding of less than 50 %.
- ☒ Both compounds have the same crystal structure and at least one of the compounds has a fraction of ionic bonding of less than 50 %.

Consider an indexed face set representation of the mesh below.



Vertices

0	1
1	1
2	1
0	0
1	0
2	0

Faces

1	4	5
?	?	?
2	3	5
5	3	6

What should the second row of the faces matrix be?

☐ [6 2 5]

☐ [2 4 5]

☐ [1 2 3]

☐ [1 4 6]

☒ [1 2 5]

☐ [6 1 3]

Copper is tested in tension. Initially the specimen is loaded to the 0.2 % proof stress. What happens when the specimen is tested further and the stress increases by a factor of 10 % (which still remains below the fracture strength)?

☒ The plastic strain increases by 10 %, while the elastic strain remains unchanged.

☐ Both strains, the plastic and the elastic strain, increase by 10 %.

☐ The elastic strain increases by 10 %, while the plastic strain increases even more.

☐ The plastic strain increases by more than 10 %, while the elastic strain remains unchanged.

Consider the digital image shown.

26	25	18	45	52	38
17	42	20	21	24	21
25	60	51	51	28	06
17	38	74	66	57	11
24	18	14	15	29	04

What is the outcome of applying a 3 x 3 median filter centered at row 2, column 3.

☐ 45

☒ 42

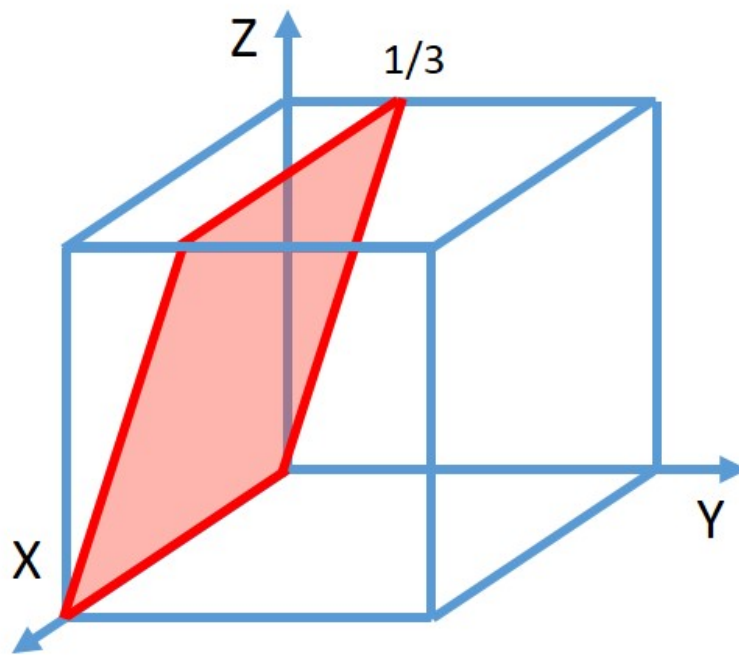
☐ 20

☐ 37

☐ 21

☐ 38

What is the proper notation of the particular crystallographic plane shown in the unit cell in the figure?



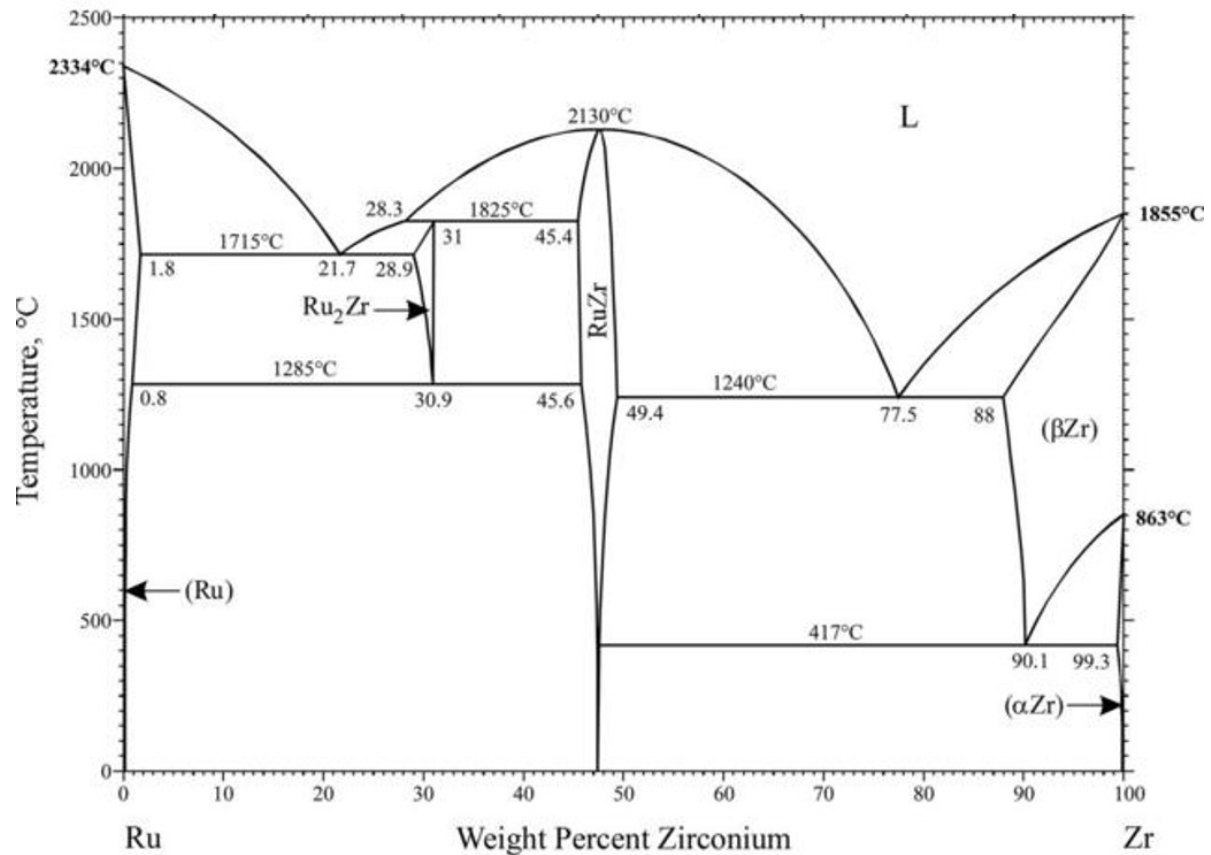
☒ $(0\bar{3}1)$

☐ $(01\bar{3})$

☐ $\{310\}$

☐ $\{0\bar{1}3\}$

The figure shows the phase diagram of the binary alloy system Ru-Zr.



A melt of a ruthenium-zirconium alloy with **77.5 wt.% Zr** is cooled slowly from 2500 °C to room temperature. The microstructure at room temperature consists of two phases. Which phases are present and how are they arranged?

- ☐ RuZr and (βZr) as alternating lamellae solely.
- ☐ RuZr and (αZr) as alternating lamellae solely.
- ☒ RuZr and (αZr) as alternating lamellae with additional (αZr).
- ☐ RuZr and (αZr) as alternating lamellae with additional RuZr.

By increasing the dislocation density in an aluminium specimen from $1 \cdot 10^{12} \text{ m}^{-2}$ to $4 \cdot 10^{14} \text{ m}^{-2}$, its yield stress increases by 230 MPa. Which additional strengthening is achieved by increasing the dislocation density even further from $4 \cdot 10^{14} \text{ m}^{-2}$ to $8 \cdot 10^{14} \text{ m}^{-2}$?

- ☐ About 460 MPa.
- ☒ About 100 MPa.
- ☐ About 330 MPa.
- ☐ About 230 MPa.

High-density polyethylene (HDPE) and low-density polyethylene (LDPE) do not only differ in their density. Which statement is correct when comparing two batches of HDPE and LDPE with same molar mass?

- ☒ HDPE has both a higher melting temperature and a higher glass transition temperature than LDPE.
- ☐ HDPE has both a lower melting temperature and a lower glass transition temperature than LDPE.
- ☐ HDPE has a higher melting temperature, but a lower glass transition temperature than LDPE.
- ☐ HDPE has a lower melting temperature, but a higher glass transition temperature than LDPE.

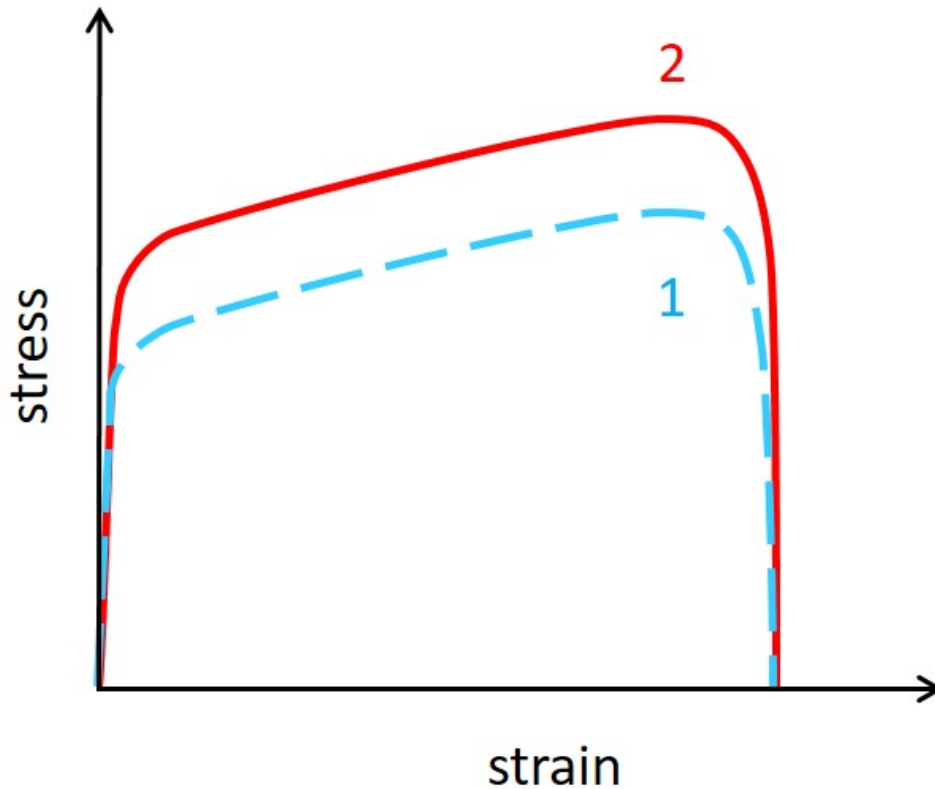
Complete miscibility between two metallic elements requires similarity of several material properties. Which of the mentioned material properties will impede complete miscibility in case of a too large difference?

- ☐ Molar mass
- ☐ Mass density
- ☐ Poisson ratio
- ☒ Electronegativity
- ☐ Electrical resistivity
- ☐ Young's modulus

A specimen of pure gold is cooled slowly from room temperature to the temperature of liquid nitrogen (-196 °C). How does this affect the defect content?

- ☒ The vacancy concentration decreases.
 - ☐ The density of grain boundaries decreases.
 - ☐ The grain size decreases.
 - ☐ The dislocation density decreases.
 - ☐ Not at all.
-

The stress strain diagram shows two flow curves obtained for two different metallic specimens (1 and 2) at room temperature under the same conditions.



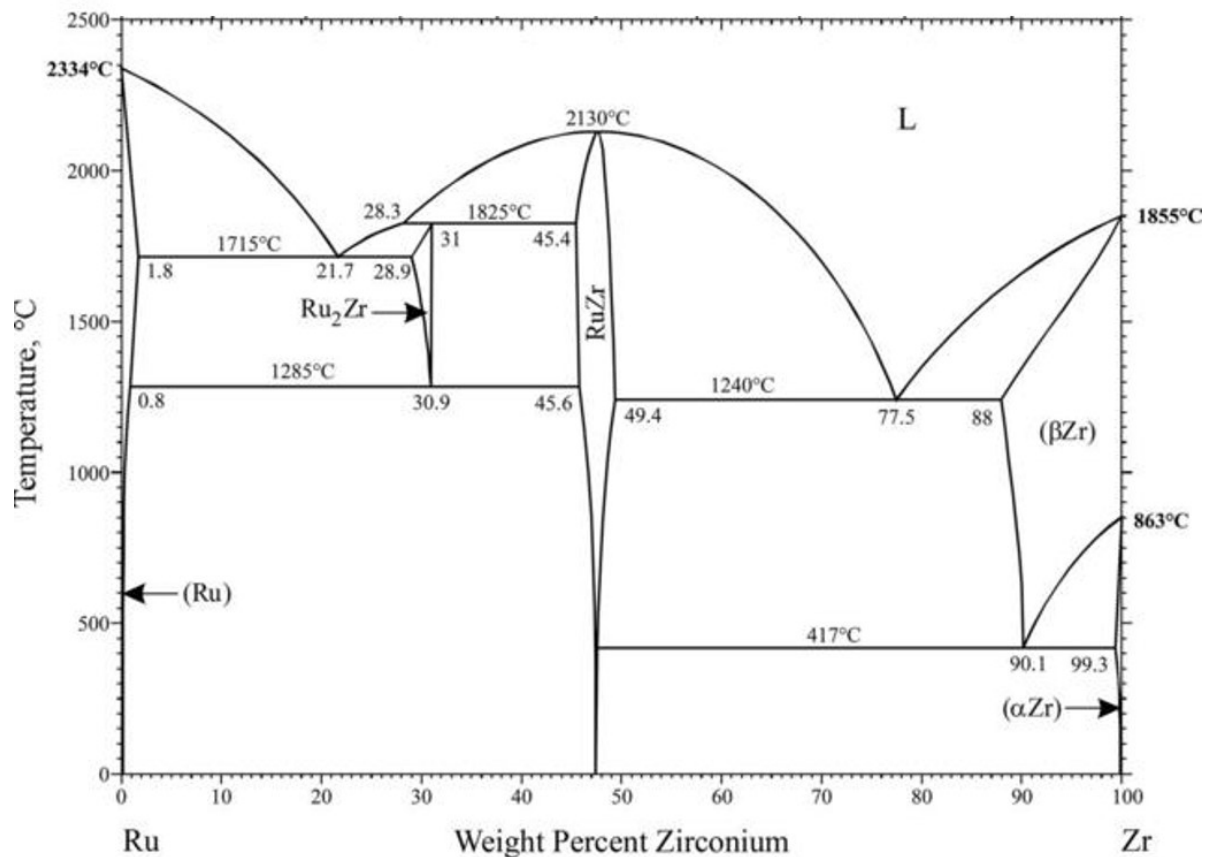
Compare tensile strength and toughness of the two metallic specimens. Which statement is correct?

- ☒ Specimen 1 has neither a higher tensile strength, nor a higher toughness than specimen 2.
 - ☐ Specimen 1 has not a higher tensile strength, but a higher toughness than specimen 2.
 - ☐ Specimen 1 has a higher tensile strength, but a lower toughness than specimen 2.
 - ☐ Specimen 1 has both a higher tensile strength and a higher toughness than specimen 2.
-

How is crack growth in metals under tension affected by their ability to deform plastically?

- ☐ Cracks grow faster when the metal can deform plastically.
- ☒ Cracks grow slower when the metal can deform plastically.
- ☐ The ability to deform plastically has no effect on crack growth.
- ☐ Cracks cannot grow at all when the metal can deform plastically.

The figure shows the phase diagram of the binary alloy system Ru-Zr.



A melt of a ruthenium-zirconium alloy with **38 wt.% Zr** is cooled slowly from 2500 °C to room temperature. What is the lowest Zr content in the melt during solidification?

- ☐ 21.7 wt.% Zr
- ☐ 38 wt.% Zr
- ☒ 28.3 wt.% Zr
- ☐ 30.9 wt.% Zr

Consider a hypothetical metal with an electron mobility of $0.002 \text{ m}^2/(\text{V}\cdot\text{s})$. Moreover, the material has a face-centered cubic structure with a lattice parameter of 0.38 nm , a density of 22.6 g/cm^3 , and a molar mass of 192 g/mol . What is the electrical conductivity assuming one conduction electron per atom?

☒ About $2.3 \cdot 10^7 \text{ S/m}$.

☐ About $6.5 \cdot 10^6 \text{ S/m}$.

☐ About $1.4 \cdot 10^3 \text{ S/m}$.

☐ About $1.4 \cdot 10^8 \text{ S/m}$.

Consider a polyethylene molecule without side branches and a molar mass of 5 kg/mol . What is the ratio between the straightened chain length and the average distance between start and end of the chain molecule (carbon-carbon bonds have a length of 0.154 nm and form an angle of 109°)?

☒ About 11.

☐ About 18.

☐ About 15.

☐ About 13.

Atoms of the element antimony crystallize under high pressure into a cubic phase with a mass density of 7.48 g/cm^3 . The molar mass of antimony is 121.7 g/mol , its atomic radius is 0.150 nm . Which crystal structure has that cubic solid phase?

☒ A diamond structure.

☐ A simple cubic structure.

☐ A face-centered cubic structure.

☐ A body-centered cubic structure.

Consider simulating the temperature distribution in a rod of metal as a 1D heat transfer problem using the heat equation. The rod is 1 m long. The two ends of the rod are defined at $x = 0$ m (point A) and $x = 1$ m (point B). In the steady state solution, the temperature in the center of the rod is 30°C . Which combination of initial temperature (in the entire rod), boundary condition at point A and boundary condition at point B could **not** have produced this result?

☒ Initial temperature 20°C , point A: $\partial u/\partial n = 0$, point B: $\partial u/\partial n = 0$

☐ Initial temperature 20°C , point A: $u=30^\circ\text{C}$, point B: $\partial u/\partial n = 0$

☐ Initial temperature 20°C , point A: $u=20^\circ\text{C}$, point B: $u=40^\circ\text{C}$

☐ Initial temperature 30°C , point A: $u=20^\circ\text{C}$, point B: $u=40^\circ\text{C}$

The ceramic material $\text{YBa}_2\text{Cu}_3\text{O}_7$ is a superconductor at low temperatures. At room temperature, $\text{YBa}_2\text{Cu}_3\text{O}_7$ has an electrical conductivity of $5 \cdot 10^5 \text{ S/m}$. Consider a wire of this material in the shape of a cylinder in an electrical field of 0.1 V/m along its axis. What must the diameter of the wire be, if it carries a current of 4 mA under these conditions at room temperature?

☐ About $1.6 \mu\text{m}$.

☒ About $320 \mu\text{m}$.

☐ About 1.6 mm .

☐ About $8 \mu\text{m}$.

Consider a 3D image of a material consisting of a solid phase and some pores. The image is segmented into solid phase and pores and the volume of the pores is determined to 0.157 mm^3 . The 3D image has $300 \times 300 \times 300$ voxels and a voxel size of $20 \mu\text{m} \times 20 \mu\text{m} \times 20 \mu\text{m}$. What is the volume fraction of the pores?

☐ 0.58

☐ 0.73

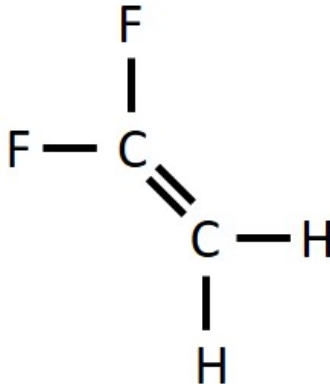
☐ 0.19

☒ 0.00073

☐ 0.00019

☐ 0.00058

The figure shows the monomer of the polymer polyvinylidene difluoride (PVDF).



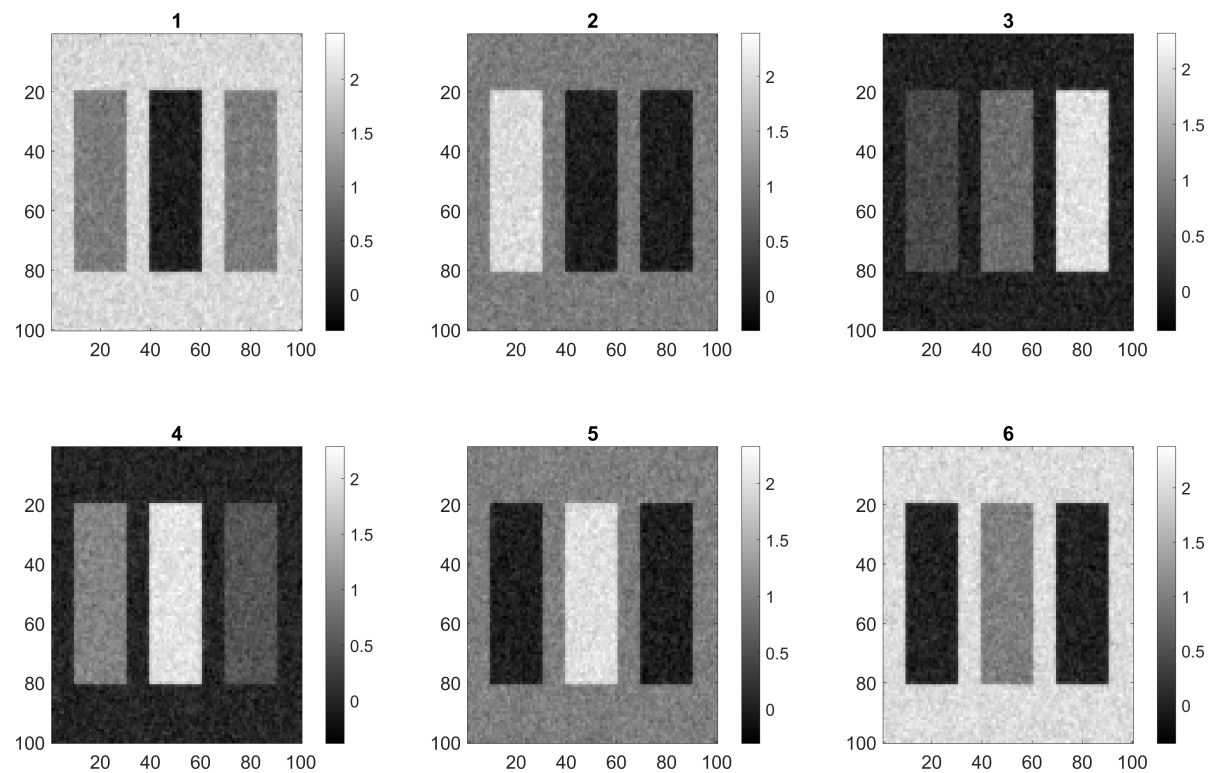
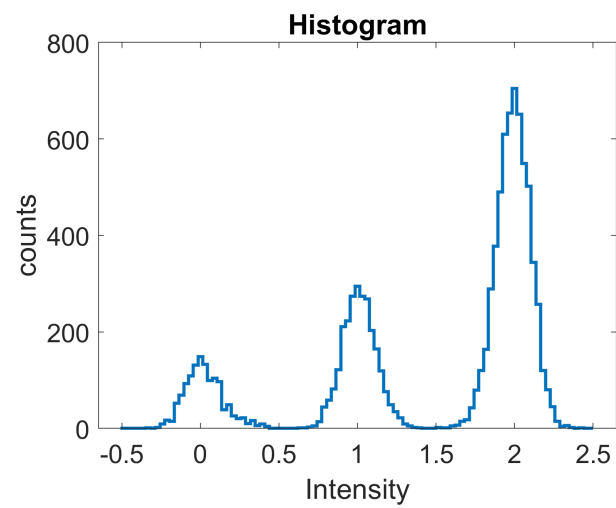
In which condition can PVDF be found?

- ☐ crystalline.
 - ☒ syndiotactic.
 - ☐ atactic.
 - ☐ semi-crystalline.
-

Which point defects have the least effect on strengthening of a metal?

- ☐ Large substitutional atoms
 - ☒ Self-interstitial atoms
 - ☐ Small interstitial atoms
 - ☐ Vacancies
-

The figure shows the histogram of one of the six images.



Which image does the histogram correspond to?

☒ 1

☐ 5

☐ 6

☐ 3

☐ 4

☐ 2

A laminate composite is manufactured from several plates made of three different metals (aluminium, copper and lead) such that their volume fraction is the same. The table summarizes mechanical properties of the three metals.

Metal	Young's modulus / GPa	Yield strength / MPa	Tensile strength / MPa	Ductility / %
Al	71.5	128	221	10
Cu	130	130	283	17
Pb	15.4	25.4	36.7	25

Which metal is the first to deform plastically when the laminate is loaded in tension along the plates?

☐ Due to the mechanical restrictions, all metals in the composite must yield simultaneously.

☐ Cu yields first.

☒ Pb yields first.

☐ Al yields first.